



Ambiguity and informativeness of (non-)trading [☆]

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ABSTRACT

We study a sequential trading mechanism with ambiguity-averse agents modeled by multiple prior preferences. Informed traders generally mix between trading and non-trading, and their trading probability decreases with ambiguity. If agents are sufficiently ambiguous, informed traders do not trade, and only noise traders place orders; trading becomes uninformative. When signal accuracy is ambiguous, trading can make public beliefs more ambiguous over time, which leads to social learning failures in the long run. Moreover, since informed traders may not trade, no-trade can also be informative when signal accuracy is asymmetric. Even with continuous action spaces, sufficiently high ambiguity stops social learning.

1. Introduction

Uncertainty pervades the financial markets and significantly shapes them. Natural disasters, economic instability, or political tensions expose traders to uncertainty. Trading volume and market information efficiency decrease when uncertainty increases. For example, asset price volatility and illiquidity spiked when Covid-19 hit (Just and Echaust, 2020; Baig et al., 2021) and when hurricane Sandy struck New York (Rehse et al., 2019). Investors seek safe assets and withdraw from the markets, which causes spreads to widen between risky and safe assets during crises such as the 1997–1998 crisis (Scholes, 2000) and the sub-prime mortgage crisis (Brunnermeier, 2009; Easley and O'Hara, 2010).

To study the effect of uncertainty on the financial market, it is useful to distinguish two types of uncertainty: risk and ambiguity. Risk refers to situations where agents know the distribution of uncertain outcomes, while ambiguity refers to situations where agents have no precise knowledge about the distribution of outcomes. The celebrated thought experiments of Ellsberg (1961) show that subjective expected utility (SEU) theory may not explain agents' behavior under ambiguity.

In this paper, we investigate how ambiguity affects asset trading and information efficiency. Our models are built upon a trading mechanism similar to that of Glosten and Milgrom (1985), where noise traders trade inelastically and randomly, informed traders trade rationally based on their private signals, and a market maker posts break-even prices based on public beliefs. Trading is sequential, with traders arriving at the market in an exogenous order. Both informed traders and the market maker have multiple prior preferences (Gilboa and Schmeidler, 1989; Casadesus-Masanell et al., 2000), capturing their aversion to ambiguity.

Our baseline model shows that ambiguity affects asset trading and information efficiency in different ways depending on its degree. To understand the main intuition, we compare three scenarios: low ambiguity, moderate ambiguity, and high ambiguity. In

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each scenario, we examine how a good signal influences an informed trader's decision to buy the asset and how the market maker sets the ask price.

In the low ambiguity scenario, an informed trader is certain to buy an asset if she receives a good signal. It ensures that buy orders reflect a high likelihood of the trader receiving a good signal, and the market maker posts a high ask price to counter the high likelihood of losses from informed traders. However, because the ambiguity is low, the informed trader's reservation prices to buy, i.e., the worst valuation of the asset, remain higher than the ask price. This scenario is akin to the classical sequential trading model with no ambiguity.

Under moderate ambiguity, an informed trader mixes between trading and non-trading. Suppose that, as in the low ambiguity scenario, the market maker posts an ask price below an informed trader's reservation price, and the trader buys the asset surely after seeing a good signal. However, this cannot be an equilibrium because it would lead to losses for the market maker. As ambiguity rises from low to moderate, the ambiguity-averse informed trader's reservation price decreases. And the fact that the ask price being even lower than the trader's reservation price implies the market maker's expected losses from trading with the informed trader would be too high after a good signal.

To prevent these losses, the market maker must raise the ask price. But setting the ask price strictly higher than the informed trader's reservation price does not work, either, in this scenario. The informed trader would never buy the asset even after seeing a good signal, and the high ask price leads to a strict positive profit per trade from only noise traders; however, competition pressures would push down the ask price. Thus, the only solution is for the market maker to set the ask price at the reservation price in this scenario, while the informed trader mixes between buying and non-trading after seeing a good signal. This mixed strategy allows the market maker to break even. As ambiguity increases, the informed trader trades less often to reflect the higher perceived cost.

When ambiguity reaches a high level, informed traders stop trading. In this scenario, the market maker cannot elicit informed traders to trade by setting appropriate prices. The lowest ask price it can set is the break-even price if only noise traders buy. But if an informed trader is very uncertain about the asset value, her worst evaluation can be lower than this price; so they would not buy. This leads to a failure of information aggregation in the market: All informed traders choose not to trade regardless of their signals. No private information is revealed in the market, and social learning gets trapped.

Then, we consider several extensions of the baseline model. First, we assume private signals have ambiguous accuracy. Informed traders' activities partially reveal these signals. Consequently, ambiguity will grow in the public beliefs, and trading tends to be noninformative in the long run. Second, we assume private signals are asymmetrically accurate: a bad signal is less informative than a good one. Informed traders trade less often with a bad signal than a good signal. This implies that no-trade is bad news: when no one trades in the market, it is more likely due to a bad signal than a good one. The asset price will decrease accordingly. Third, we show that the social learning failure result holds even if traders have continuous action spaces.

There is a large strand of literature that examine the effect of ambiguity aversion in the financial market, and our work also contributes to it. Ambiguity aversion influences traders' willingness to trade and can cause portfolio inertia or no-trade (Dow and da Costa Werlang, 1992). This can result in price indeterminacy (Epstein and Wang, 1994), under-diversification (Maccheroni et al., 2013), market incompleteness (Mukerji and Tallon, 2001), low liquidity (Ozsoylev and Werner, 2011), and excess volatility (Illeditsch, 2011). Furthermore, ambiguity aversion has been proposed as a possible solution to the equity premium puzzle (Chen and Epstein, 2002; Trojani and Vanini, 2002; Ju and Miao, 2012). For comprehensive insights into the influence of ambiguity on trading decisions and market dynamics, Epstein and Schneider (2010) and Guidolin and Rinaldi (2012) provide thorough overviews.

This paper studies how ambiguity affects the trading behavior of ambiguity-averse traders in a sequential trading mechanism and finds several novel results. We show that ambiguity-averse traders trade less when there is more ambiguity; this can lead to social learning and information aggregation failures, even with endogenous prices and continuous actions. Chari and Kehoe (2004) summarize the literature and point out that informational cascade results (Bikhchandani et al., 1992; Banerjee, 1992) are subject to *price critique* and *continuous investment critique*. That is, these failures disappear if prices are endogenous (Avery and Zemsky, 1998) or investment decisions are continuous (Lee, 1993). In the literature, there are attempts to overturn one or both critiques by introducing endogenous timing (Chari and Kehoe, 2004), transaction costs (Lee, 1998), exogenous gains and loss from trade (Cipriani and Guarino, 2008), and risk-averse traders (Decamps and Lovo, 2006). Our paper is unique in considering ambiguity aversion as a source of these failures, which can overcome both critiques.

Our results also contribute to the literature in market microstructure. We find ambiguity-averse traders generally mix between trading and non-trading, which echoes the findings of de Meyer and Saley (2003). However, we consider that informed traders' randomizing between trading and non-trading emerges under ambiguity aversion¹ and show that, consequently, ambiguity aversion can be an explanation for informativeness of no-trade. Moreover, our study goes beyond the classical microstructure literature, such as Glosten and Milgrom (1985) and Kyle (1985), which typically focuses on how prices eventually aggregate and reveal private information. Instead, our results suggest that the opposite could happen if traders exhibit ambiguity aversion.²

Another contribution of our paper is to study belief updating and price adjustment with ambiguity-averse agents. This distinguishes our paper from other studies on the effect of ambiguity. For example, Easley and O'Hara (2010) use Bewley's (2002) model of

¹ There is a strand of literature discussing how randomization interacts with ambiguity. As noted by some authors (Raiffa, 1961; Kuzmics, 2017), randomization can eliminate the effect of ambiguity if agents evaluate buy and sell orders with the same belief in their belief set. However, in our model, agents evaluate buy and sell orders with different worst case beliefs, where randomization cannot eliminate the effect of ambiguity (Saito, 2015; Oechssler and Roomets, 2015; Ke and Zhang, 2020). Instead, informed traders randomize since the market make post prices make them indifferent between trading and non-trading in equilibrium.

² Ford et al. (2013) and Dong et al. (2010) also explore the effect of ambiguity in the sequential trading models and find herding behaviors emerge in equilibrium. However, their notion of herding follows from Avery and Zemsky (1998), which does not prevent social learning and information aggregation in the long run.

ambiguity aversion to study how ambiguity leads to a market freeze. However, their analysis is limited to a one-shot trading situation, while our paper shows that social learning failure can persist over time and that learning with ambiguous signal accuracy results in long-term information aggregation failure. Furthermore, Condie and Ganguli (2011; 2017) consider the effect of ambiguity-aversion in a rational expectation equilibrium and find results similar to ours, such as the revelation of private information through no-trade. They show that ambiguity-averse traders' demand remains unchanged with certain private information, leading to only partially revealing prices in the absence of noise traders. In contrast, our study emphasizes the informativeness of no-trade, highlighting that market prices can adjust even when no trading activity occurs.

Our results are supported by empirical evidence in the literature. Dimmock et al. (2016a) and Dimmock et al. (2016b) finds that ambiguity averse agents reduce stock market participation, especially during the financial crisis. Kostopoulos et al. (2022) and Ben-Rephael and Izhakian (2020) find that ambiguity reduces trading and exposure to uncertain assets using individual and firm-level data. They also find that ambiguity increases book-depth and lowers price impact, consistent with our model predictions. Some studies, e.g., Hong and Li (2018) and Gao et al. (2022), find that insiders' non-activity reveals their private information in an uncertain environment. Our results of informativeness of no-trade can explain these findings.

The paper is organized as follows. Section 2 describes the baseline model and fully characterizes the equilibrium. Then, Section 3 considers several extensions of the baseline model. We first provide a general model in Section 3.1, based on which we show that (i) the market tends to an informational cascade if private signals are ambiguous in Section 3.2 and that (ii) no-trade can be informative when signals are symmetric in Section 3.3. Section 3.4 extends the baseline model with continuous action spaces. Finally, Section 4 concludes. All the proofs are collected in Appendix.

2. The baseline model

2.1. Model set-up

We start by describing the baseline model. There are three types of agents, a market maker, informed traders, and noise traders, in the model. Time is discrete and denoted by $t = 1, 2, \dots, T$ with $T \leq \infty$. There is an asset with an uncertain payoff $v \in \{0, 1\}$, which is not publicly observable until period T .

In each period t , the market maker first posts an ask price a_t and a bid price b_t . Then either an informed trader or a noise trader arrives with probability $\eta \in (0, 1)$ or $1 - \eta$, respectively. Each trader can either buy or sell one unit of the asset or choose not to trade.³ A trader's order is recorded as $z_t \in \{-1, 0, 1\}$, whereas $z_t = -1$ stands for a sell order, $z_t = 0$ for not to trade, and $z_t = 1$ for a buy order. Each trader can trade only once and leave the market forever. Noise traders are assumed to trade for exogenous reasons with inelastic demand. Their actions are randomly chosen with equal probabilities.

The *public history* or, simply, *history* h_t observable to all the agents has length $t - 1$ and contains the sequence of prices $\{a_\tau, b_\tau\}_{\tau=1}^{t-1}$ and orders $\{z_\tau\}_{\tau=1}^{t-1}$, i.e., $h_t = \{a_\tau, b_\tau, z_\tau\}_{\tau=1}^{t-1}$. H_t denotes the set of all possible histories with length $t - 1$, and as a convention, history with length 0 is an empty set, i.e., $h_1 = \emptyset$.

A public belief about the uncertain asset payoff can be summarized by $\pi_t = \Pr(v = 1 | h_t) \in [0, 1]$ in each period t . Agents can be ambiguous about the asset payoff: there can be uncertainty in their beliefs. Instead of assuming that they have a unique public belief, they have a common set $\Pi_t(h_t) \subseteq [0, 1]$, or simply Π_t , of beliefs after history h_t . We assume that Π_t is closed and convex, and let $\underline{\pi}_t = \min \Pi_t$ and $\bar{\pi}_t = \max \Pi_t$ be its lower and upper bounds, respectively. Then the set Π_t of public beliefs can be written as an interval $[\underline{\pi}_t, \bar{\pi}_t]$.

Both the market maker and informed traders are risk-neutral. However, they are ambiguity-averse, and their preferences are represented by maxmin expected utility. To set ask and bid prices in period t , the market maker wants to maximize the *min-expected profits*, $\min_{\pi_t} E^{\pi_t}[(a_t - v)I_{\{z_t=1\}} | h_t]$ and $\min_{\pi_t} E^{\pi_t}[(v - b_t)I_{\{z_t=-1\}} | h_t]$, respectively. However, we assume that the market maker can only earn zero profit for each trade because of competition pressure in the background, as similarly argued by Glosten and Milgrom (1985). Consider that there are two market makers who have the same information and face the same potential traders. Both have to set ask prices no less than $E^{\hat{\pi}_t}[v | h_t, z_t = 1]$, where $\hat{\pi}_t \in \arg \max_{\pi_t} E^{\pi_t}[v | h_t, z_t = 1]$; otherwise they would make a loss from a buy trade. Suppose that the first one sets an ask price a_t^1 so that $a_t^1 > E^{\hat{\pi}_t}[v | h_t, z_t = 1]$. Then the second market maker can undercut the first one by setting a price a_t^2 such that $E^{\hat{\pi}_t}[v | h_t, z_t = 1] < a_t^2 < a_t^1$. The Bertrand competition leads to zero profit in equilibrium in period t , satisfying

$$a_t = \max_{\pi_t \in \Pi_t} E^{\pi_t}[v | h_t, z_t = 1], \quad (2.1)$$

$$b_t = \min_{\pi_t \in \Pi_t} E^{\pi_t}[v | h_t, z_t = -1]. \quad (2.2)$$

Let $x = (a_t, b_t)_{t=1}^T$ denote a pricing rule of the market maker.

Without ambiguity, zero profit conditions result in that the ask and bid prices are the expected value of the asset conditional on a buy order and a sell order, respectively, in period t . However, these conditions are slightly modified under ambiguity; ask and bid prices rely on different beliefs. The intuition is that, because of uncertainty aversion, when there is a buy order, the market maker

³ To clarify, “buy” and “sell” orders are defined from a trader's perspective.

wants to sell the asset at the highest conditional expected value, and, when there is a sell order, the market maker wants to buy the asset at the lowest conditional expected value.

The informed trader who meets the market maker in period t receives a conditionally independent *private signal* $s_t \in \{0, 1\}$. In the baseline model, we assume that the conditional probabilities of private signals are as follows:

$$\Pr(s_t = 1|v = 1) = \Pr(s_t = 0|v = 0) = q \in (1/2, 1). \quad (2.3)$$

We simply refer $s_t = 1$ a good signal and $s_t = 0$ a bad signal. The parameter q captures the signal accuracy, and the assumption that $q > 1/2$ implies that signals are sufficiently informative. Observing a good signal indicates a higher conditional probability of a valuable asset. Moreover, the assumption also states that the accuracy of private signals is unambiguous and symmetric.

Observing new information, agents update the set of prior beliefs in a full Bayesian way (Pires, 2002; Epstein and Schneider, 2003); that is, all the beliefs in the set of prior beliefs are updated by Bayes rule. Given a closed and convex set Π_t of public beliefs, let $\pi_t^{(s_t)}$ denote the *updated belief* given prior belief $\pi_t \in \Pi_t$ for an informed trader observing private signal s_t . Particularly,

$$\begin{aligned} \pi_t^{(1)} &= \frac{\pi_t q}{\pi_t q + (1 - \pi_t)(1 - q)}, \\ \pi_t^{(0)} &= \frac{\pi_t(1 - q)}{\pi_t(1 - q) + (1 - \pi_t)q}, \end{aligned}$$

for any belief $\pi_t \in \Pi_t$. The set of the informed trader's posterior beliefs, $\Pi_t^{(s_t)}$, in period t , is obtained via Bayes rule in a prior-by-prior manner, i.e.,

$$\Pi_t^{(s_t)} = \left\{ \pi_t^{(s_t)} \mid \pi_t \in \Pi_t \right\}.$$

Note that $\pi_t^{(s_t)}$ is strictly increasing in π_t ; hence, we have that $\min \Pi_t^{(s_t)} = \underline{\pi}_t^{(s_t)}$ and $\max \Pi_t^{(s_t)} = \bar{\pi}_t^{(s_t)}$. Therefore, $\Pi_t^{(s_t)} = [\underline{\pi}_t^{(s_t)}, \bar{\pi}_t^{(s_t)}]$.

The informed trader chooses a mixed strategy $\sigma_t : H_t \times \mathbb{R}_+^2 \times S \rightarrow \Delta$, where $\sigma_t = (\zeta_t, \iota_t)$ with ζ_t and ι_t being the probabilities of buying and selling, respectively.⁴ Denote Σ_t the set of all the possible trading strategies of the informed trader in period t . The informed trader chooses her trading strategy $\sigma_t \in \Sigma_t$ to maximize her min-expected profit,

$$\min_{\pi_t \in \Pi_t} E^{\pi_t} [\zeta_t(v - a_t) + \iota_t(b_t - v) \mid h_t, a_t, b_t, s_t]. \quad (2.4)$$

The min-expected profit linearly depends on the trading probabilities, ζ_t and ι_t . Buying or selling the asset is preferred and played with positive probability only if it is at least as good as the other two actions. Thus, it is straightforward to see that the optimal trading strategies stated in the following lemma.

Lemma 2.1. *Let the set of public beliefs be Π_t after history h_t . For an informed trader who observes private signal s_t , her optimal trading strategies satisfy the following:*

$$\begin{aligned} \zeta_t(h_t, a_t, b_t, s_t) \geq 0 &\iff \underline{\pi}_t^{(s_t)} - a_t \geq \max \left\{ 0, b_t - \bar{\pi}_t^{(s_t)} \right\}, \\ \iota_t(h_t, a_t, b_t, s_t) \geq 0 &\iff b_t - \bar{\pi}_t^{(s_t)} \geq \max \left\{ 0, \underline{\pi}_t^{(s_t)} - a_t \right\}. \end{aligned}$$

Furthermore, if $a_t \geq b_t$, then the above conditions are reduced to

$$\begin{aligned} \zeta_t(h_t, a_t, b_t, s_t) \geq 0 &\iff \underline{\pi}_t^{(s_t)} \geq a_t, \\ \iota_t(h_t, a_t, b_t, s_t) \geq 0 &\iff \bar{\pi}_t^{(s_t)} \leq b_t. \end{aligned}$$

Lemma 2.1 shows that ambiguity-averse traders decide to buy or sell depending on different beliefs. Whether to buy the asset depends on the worst updated belief $\underline{\pi}_t^{(s_t)}$, while selling the asset is determined by the best updated belief $\bar{\pi}_t^{(s_t)}$. Let $\sigma = (\sigma_t)_{t=1}^T$ denote a trading strategy.

Our solution concept is similar to a perfect Bayesian equilibrium, which consists of a belief system and a strategy profile, and defined as follows:

Definition 2.1 (Equilibrium). An equilibrium of the trading mechanism in the baseline model consists of a pricing rule x , a trading strategy σ , and a belief system $\Pi = (\Pi_t)_{t=1}^T$ such that: (i) The belief system Π is consistent w.r.t. the prior belief set Π_1 , the pricing rule x , and the trading strategy σ ; that is, every $\pi_t \in \Pi_t$ is obtained by Bayesian updating wherever possible, given (a_t, b_t, σ_t) and some prior belief $\pi_1 \in \Pi_1$. In every period t , $t = 1, \dots, T$, (ii) prices, a_t and b_t , satisfy zero profit conditions (2.1) and (2.2) given the belief system Π and the trading strategy σ_t , and (iii) trading strategy σ_t maximizes an informed trader's min-expected profit (2.4) given the belief system Π and prices, a_t and b_t .

⁴ Here $\Delta = \{(\zeta, \iota) \in [0, 1] \times [0, 1] : \zeta + \iota \leq 1\}$, which is homeomorphic to a simplex in $\mathbb{R}^3$.

Note that the usual definition of belief consistency is extended to accommodate a system of belief sets. Given a pricing rule x and a trading strategy σ , a consistent belief system can be constructed inductively as follows: Suppose that the set Π_t of public beliefs and $h_{t+1} = (h_t, a_t, b_t, z_t)$ is given. Then, the set of public beliefs in period $t + 1$ is $\Pi_{t+1} = \{\pi_{t+1} = \Pr^{x, \sigma, \pi_t}(v = 1 | h_{t+1}) | \pi_t \in \Pi_t\}$, where $\Pr^{x, \sigma, \pi_t}(\cdot | I_t)$ is the probability conditional on event I_t given a pricing rule x , a trading strategy σ , and a prior belief π . We will discuss the belief updating in more depth in Section 2.3.

2.2. Equilibrium trading strategy and prices

In this subsection, we consider an equilibrium (x, σ, Π) and derive the necessary conditions for the trading strategy and compute the prices in the equilibrium. Let us fix a history h_t and its corresponding public belief Π_t . The first lemma in the following shows that an informed trader must neither buy upon a bad signal nor sell upon a good signal.⁵ To see it, suppose that an informed trader buys the asset when seeing a bad signal. It must be that $a_t \leq \underline{\pi}_t^{(0)}$ by Lemma 2.1. If it is the case, an informed trader must also buy the asset when seeing a good signal. Thus, $z_t = 1$ is a signal that indicates a higher likelihood of $v = 1$ than an informed trader's signal $s_t = 0$. Then the zero profit condition (2.1) implies that $a_t > \bar{\pi}_t^{(0)}$, which is a contradiction since $\underline{\pi}_t^{(0)} \leq \bar{\pi}_t^{(0)}$.

Lemma 2.2. *In any equilibrium, if the market maker posts prices, a_t and b_t , satisfying zero profit conditions, then it must hold that*

$$\zeta_t(h_t, a_t, b_t, 0) = 0, \quad \zeta_t(h_t, a_t, b_t, 1) \geq 0,$$

$$i_t(h_t, a_t, b_t, 0) \geq 0, \quad i_t(h_t, a_t, b_t, 1) = 0.$$

By Lemma 2.2, we can primarily analyze an informed trader's probabilities of buying upon a good signal, $\zeta_t(h_t, a_t, b_t, 1)$, and selling upon a bad signal, $i_t(h_t, a_t, b_t, 0)$, henceforth simply denoted as $\hat{\zeta}_t$ and $\hat{i}_t$. It is worth noting that in the absence of ambiguity, we could derive a stronger result: informed traders would always trade after observing their signals, i.e., $\hat{\zeta}_t = \hat{i}_t = 1$. However, as the subsequent analysis will show, with ambiguity aversion the informed trader's decisions depend on the set of public beliefs and mixed strategies can arise in equilibrium.

It turns out that an informed trader's decision is characterized by the set of public beliefs. Let a set Π_t of public beliefs be given, and recall the best belief $\bar{\pi}_t = \max \Pi_t$ and worst belief $\underline{\pi}_t = \min \Pi_t$. We propose that $\delta_t := \bar{\pi}_t(1 - \underline{\pi}_t)/\underline{\pi}_t(1 - \bar{\pi}_t)$ measures the *level of ambiguity* in the public belief, which is a measure of the discrepancy between best and worst beliefs. Moreover, we define two thresholds, $\theta_*(q)$ and $\theta^*(q)$, depending on the signal accuracy, as follows:

$$\theta_*(q) := \frac{q}{1-q} \frac{1-q+\gamma}{q+\gamma} \text{ and } \theta^*(q) := \frac{q}{1-q},$$

where $\gamma = (1-\eta)/3\eta$. Note that $\theta_*(q) < \theta^*(q)$ since $1-q+\gamma/q+\gamma < 1$.

The following proposition shows that there are three patterns of an informed trader's trading strategy depending on the level of ambiguity: If the level of ambiguity is low, an informed trader trades with certainty. If the level of ambiguity is high, an informed trader never trades. However, when the level of ambiguity is moderate, an informed trader mixes between trading and non-trading.

Proposition 2.1. *In any equilibrium, given a history h_t and the set Π_t of public beliefs, the following holds:*

- (i) *An informed trader trades with certainty, i.e., $\hat{\zeta}_t = \hat{i}_t = 1$, if and only if $\delta_t \leq \theta_*(q)$; in this case, ask and bid prices, respectively, are $a_t = a_t^*$ and $b_t = b_t^*$ where*

$$a_t^* := \frac{\bar{\pi}_t(q+\gamma)}{\bar{\pi}_t(q+\gamma) + (1-\bar{\pi}_t)(1-q+\gamma)}, \quad (2.5)$$

$$b_t^* := \frac{\underline{\pi}_t(1-q+\gamma)}{\underline{\pi}_t(1-q+\gamma) + (1-\underline{\pi}_t)(q+\gamma)}. \quad (2.6)$$

- (ii) *There exists a unique $p_t \in (0, 1)$ such that an informed trader trades with probability p_t , i.e., $\hat{\zeta}_t = \hat{i}_t = p_t$, if and only if*

$$\theta_*(q) < \delta_t < \theta^*(q); \quad (2.7)$$

in this case, ask and bid prices are $a_t = \underline{\pi}_t^{(1)}$ and $b_t = \bar{\pi}_t^{(0)}$, respectively.

- (iii) *An informed trader does not trade, i.e., $\hat{\zeta}_t = \hat{i}_t = 0$, if and only if $\delta_t \geq \theta^*(q)$; in this case, ask and bid prices are $a_t = \bar{\pi}_t$ and $b_t = \underline{\pi}_t$, respectively.*

Note that, if there is no ambiguity, informed traders always trade, which is a special case of part (i) in the proposition. If Π_t is a singleton, i.e., no ambiguity, implies that $\delta_t = 1$. However, $\theta_*(q)$ is greater than one. Thus, part (i) indicates that ambiguity has no substantial effect on traders' behavior when it is sufficiently small. In fact, the inequality $\delta_t \leq \theta_*(q)$ is equivalent to that both $\underline{\pi}_t^{(1)} \geq a_t^*$

⁵ Lemma 2.2 directly follows Theorem 3.2 in Section 3.1.

and $\bar{\pi}_t^{(0)} \leq b_t^*$ hold, where the prices a_t^* and b_t^* satisfy the zero profit conditions given that an informed trader buys the asset upon a good signal and sells the asset upon a bad signal.

However, when the level of ambiguity surpasses the threshold $\theta_*(q)$, ambiguity takes effect. Suppose, for contradiction, that an informed trader still buys with certainty in this case. Expecting that, the market maker would post an ask price a_t^* as defined in equation (2.5). However, the level of ambiguity is too high for an informed trader to buy; in fact, inequality (2.7) is equivalent to $\bar{\pi}_t < \underline{\pi}_t^{(1)} < a_t^*$.⁶ Knowing that an informed trader will not buy the asset with an ask price a_t^* , the market maker has to lower the ask price, equating it with an informed trader's reservation price conditional on a good signal, i.e., $\underline{\pi}_t^{(1)}$.

Thus, an informed trade is indifferent between buying and no-trade upon a good signal and mixes between trading or not with a trading probability $p_t \in (0, 1)$.⁷ Then the observed order z_t becomes a sufficiently noisy observation of traders' private information so that the market maker breaks even by posting an ask price exactly equal to $\underline{\pi}_t^{(1)}$. Moreover, as the level of ambiguity increases, an informed trader's reservation price decreases; hence, she reduces the probability of buying, and it makes trading orders noisier. Corollary 2.1 summarizes this result.

Corollary 2.1. Consider a history h_t and the set Π_t of public beliefs such that inequality (2.7) holds, i.e., part (ii) of Proposition 2.1. Then, the trading probability p_t strictly decreases with the level of ambiguity δ_t .

If the level of ambiguity gets even higher and surpasses the threshold $\theta^*(q)$, an informed trader's trading probability hits the zero lower bound. The market maker cannot post an ask price lower than $\bar{\pi}_t$ and a bid price higher than $\underline{\pi}_t$; otherwise, it makes losses by trading with noise traders. However, the lowest ask price and the highest bid price the market maker can post fall into the no-trade region of an informed trader.⁸ An informed trader finds it never optimal to trade under this situation. All the buy and sell orders come from noise traders and convey no information about informed traders' private signals. Consequently, the market maker posts prices relying only on public beliefs.

The informed traders do not trade when ambiguity is sufficiently high, which is reminiscent of the no-trade result or portfolio inertia under ambiguity when prices are exogenous (Dow and da Costa Werlang, 1992). However, prices are endogenously determined in our sequential trading mechanism. When there is only a little ambiguity, the market maker can set a sufficiently large ask–bid spread beyond the no-trade region so that an informed trader is willing to (partially) reveal the private information. However, the market maker fails to do so when ambiguity is too high: By posting an ask price equal to $\underline{\pi}_t^{(1)}$ (or lower), the expected loss from informed traders cannot be compensated by the profit from noise traders in the market maker's worst belief. Hence, the market maker sets an ask (bid) price that breaks even with noise traders and is higher than (lower) than an informed trader's reservation price to buy (sell).

Note that, with the ambiguity-averse market maker, the ask–bid spreads must be strictly positive. In the cases where informed traders trade with positive probabilities, each order z_t is a noisy observation of an informed trader's private signal. A buy order acts as a good signal and a sell order as a bad signal. Thus, the ask price must be strictly higher than the bid price given the same prior belief. Moreover, because of ambiguity aversion, the ask price is set according to the upper bound of the public beliefs, and the bid price is set by the lower bound, which further enlarges the ask–bid spreads. In the case where informed traders do not trade, the set of public beliefs must not be a singleton. The ask price is simply the best prior belief, and the bid price is the worst prior belief.

How does ambiguity affect the ask–bid spread? To answer this question, consider two sets, Π_t and Π'_t , of public belief, and Π'_t is said to be *more ambiguous* than Π_t if $\Pi_t \subset \Pi'_t$. This relation is consistent with the level of ambiguity. Let δ_t and δ'_t be the level of ambiguity in Π_t and Π'_t , respectively. Then that Π'_t is more ambiguous than Π_t implies that $\delta'_t > \delta_t$. The following result shows that, loosely speaking, the ask–bid spread increases with ambiguity.

Proposition 2.2. Let (a_t, b_t) and (a'_t, b'_t) be the equilibrium ask and bid prices as described in Proposition 2.1 corresponding to the beliefs Π_t and Π'_t , respectively. If Π'_t is more ambiguous than Π_t , then $a'_t - b'_t > a_t - b_t$.

When the market maker is ambiguity-averse, it will push up the ask price and press down the bid price to compensate for its aversion to ambiguity. The more ambiguity there is in the market, the larger the ask–bid spread is. Hence, the market maker's ambiguity aversion reinforces the problem of adverse selection, and more uncertainty induces larger ask–bid spreads.

2.3. Belief updating and equilibrium

To specify a consistent belief system, our analysis focuses only on *equilibrium paths*, i.e., the histories reachable in an equilibrium. In equilibrium, the publicly observed trading orders z_t can also be considered as a public signal with conditional distributions

⁶ Inequality (2.7) is also equivalent to $b_t^* < \bar{\pi}_t^{(0)} < \underline{\pi}_t$.

⁷ In our analysis, it is not the case that informed traders evaluate buying and selling with the same worst case belief and eliminate ambiguity through randomization (Raiffa, 1961; Kuzmics, 2017). Instead, we consider that informed traders evaluate buying and selling under different worst case beliefs. Theoretical analysis shows that randomization cannot eliminate the effect of ambiguity (Saito, 2015; Ke and Zhang, 2020), which is substantiated in some experiments (Oechssler and Roomets, 2015). In our model, when the level of ambiguity falls into $(\theta_*(q), \theta^*(q))$, the market maker posts an ask (bid) price that equates an informed trader's worst belief to evaluate a buy (sell) transaction with a good (bad) signal. Therefore, informed traders are indifferent between buying (selling) and non-trading, of which mixing is also optimal, when seeing a good (bad) signal.

⁸ The inequality $\delta_t \geq \theta^*(q)$ is equivalent to $\underline{\pi}_t^{(1)} \leq \bar{\pi}_t$ and $\bar{\pi}_t^{(0)} \geq \underline{\pi}_t$.

$\Pr(z_t|h_t, v)$.⁹ Let $l_t(z_t) = \Pr(z_t|h_t, v=1)/\Pr(z_t|h_t, v=0)$ be the *likelihood ratio* function. In a history h_t on an equilibrium path, a public belief π_{t+1} is obtained by Bayes rule given a prior belief $\pi_t \in \Pi_t$ if and only if

$$\frac{\pi_{t+1}}{1 - \pi_{t+1}} = \frac{\pi_t}{1 - \pi_t} l_t(z_t)$$

for any $z_t \in \{-1, 0, 1\}$. Let a history h_t on an equilibrium path and the corresponding set Π_t of public beliefs be given, and ask and bid prices, a_t and b_t , are as specified in Proposition 2.1. After the trading order z_t is observed, the set Π_{t+1} of public beliefs in the subsequent history $h_{t+1} = (h_t, a_t, b_t, z_t)$ on the equilibrium path can be characterized as follows:

$$\Pi_{t+1} = \left\{ \pi_{t+1} \in (0, 1) \mid \exists \pi_t \in \Pi_t : \frac{\pi_{t+1}}{1 - \pi_{t+1}} = \frac{\pi_t}{1 - \pi_t} l_t(z_t) \right\}.$$

This characterization of a consistent belief system gives us the following result.

Proposition 2.3. *In any equilibrium, for all the periods $t, t = 1, \dots, T$, the level of ambiguity remains the same, i.e., $\delta_t = \delta_1$, on the equilibrium path.*

In the baseline model, it turns out that the level of ambiguity is constant along an equilibrium path. This result greatly helps us characterize equilibrium outcomes, i.e., the trading strategies and pricing rules on equilibrium paths, in the baseline model. In any equilibrium, the market maker's pricing rule and an informed trader's decision can always be described by one of the three possible cases in Proposition 2.1. Then Proposition 2.4 follows Proposition 2.2 and 2.3 immediately.

Proposition 2.4. *One of the following must hold in any equilibrium.*

- (i) If $\delta_1 \leq \theta_*(q)$, an informed trader trades with probability one, and ask and bid prices are a_t^* and b_t^* , respectively, as defined in equation (2.5) and (2.6) for all $t \geq 1$ on equilibrium path.
- (ii) If $\theta_*(q) < \delta_1 < \theta^*(q)$, an informed trader trades with probability $p \in (0, 1)$, and ask and bid prices are $\bar{\pi}_t^{(1)}$ and $\bar{\pi}_t^{(0)}$, respectively, for all $t \geq 1$ on equilibrium path.¹⁰
- (iii) If $\delta_1 \geq \theta^*(q)$, an informed trader never trades, and ask and bid prices are $\bar{\pi}_t$ and $\underline{\pi}_t$, respectively, for all $t \geq 1$ on equilibrium path.

In the first two cases, where informed traders trade with positive probabilities, trading orders act as informative signals for the market maker. Therefore, we can extend the Proposition 4 of Glosten and Milgrom (1985) and the Proposition 4 of Avery and Zemsky (1998) when the ambiguity is sufficiently small. When $\delta_1 < \theta^*(q)$, ask and bid prices converge to the true asset value as $T \rightarrow \infty$ almost surely, and thus ask–bid spreads converge to zero. The set Π_t of public beliefs also shrinks in the sense that $\bar{\pi}_t - \underline{\pi}_t$ converges to zero almost surely.

Fig. 2.1 and 2.2 plot two simulated equilibrium paths of case (i) and (ii), respectively. When informed traders trade certainly, Fig. 2.1 shows that convergence is achieved fairly fast. It only takes several hundreds of periods to make the initial prices around 0.5 converge to the true asset value. However, it may take an extraordinarily long time to converge when informed traders mix between trading and non-trading, even though convergence is predicted by theory. In the sample path depicted in Fig. 2.2, there is no signs of convergence even after 1000 periods.

In the third case, all the informed traders do not trade, and this decision is independent of her private signal. We call such a state a *noninformative trading regime*: all the trading becomes uninformative. The state is observationally equivalent to an informational cascade where social learning stops and information aggregation fails in the long run.¹¹ The ask and bid prices are constant, and the ask–bid spreads never converge. The study of Avery and Zemsky (1998) emphasizes a “price critique” on the possibility of information aggregation failure in the sequential trading mechanism à la Glosten and Milgrom (1985) without ambiguity: the competitive market maker can adjust the price aligning with the public belief; therefore, traders’ optimal response always partially reveals their private signals. However, our result shows such a price critique can be overturned by introducing ambiguity aversion.

2.4. Ambiguity-neutral market maker

In this subsection, we consider a *variant model* where the market maker is assumed to be ambiguity-neutral, in contrast to the ambiguity-averse market maker analyzed thus far. We show that the results are qualitatively the same as the baseline model when the market maker has different preferences towards ambiguity. Thus, our key findings in the baseline model are robust to the assumption of the market maker's ambiguity preferences.

⁹ Since that ask and bid prices are uniquely determined as in Proposition 2.1 if a history h_t and the corresponding set Π_t of public beliefs are given, $\Pr(z_t|h_t, v) = \Pr(z_t|h_t, a_t, b_t, v)$ on equilibrium paths.

¹⁰ We say that an informed trader *trades with probability p* if she buys with probability p upon a good signal and sells with probability p upon a bad signal.

¹¹ The traders’ herding on no-trade in our model is not due to that agents’ beliefs are overwhelmed by the public belief, as the informational cascade in Bikhchandani et al. (1992). However, when informed traders herd on no-trade, their actions are uninformative and prevent further social learning as prescribed in an informational cascade.

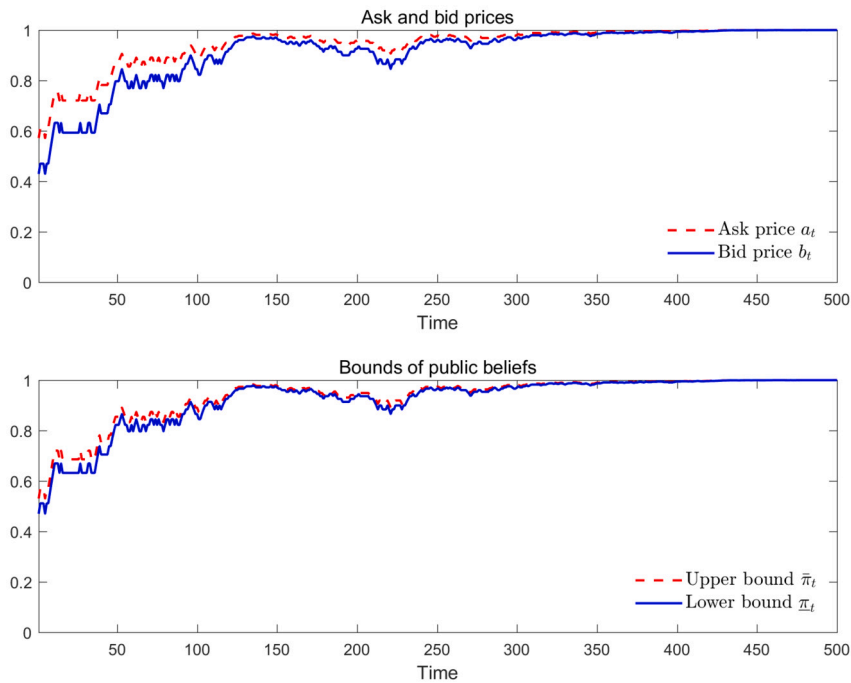


Fig. 2.1. Baseline model with the following parameters: $\nu = 1$, $\eta = 0.5$, $\underline{\pi}_1 = 0.47$, $\bar{\pi}_1 = 0.53$, and $q = 0.6$.

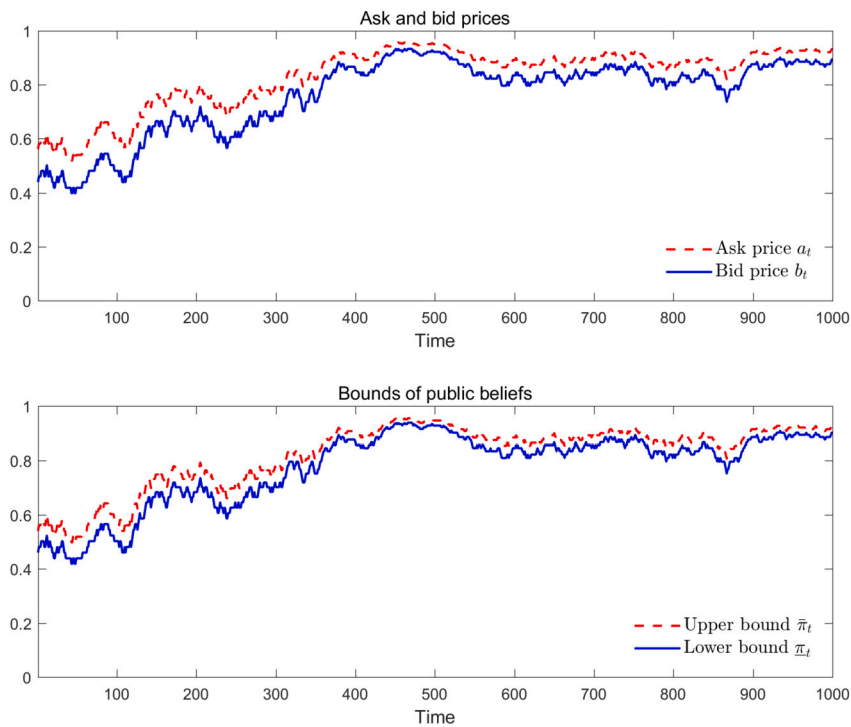


Fig. 2.2. Baseline model with the following parameters: $\nu = 1$, $\eta = 0.5$, $\underline{\pi}_1 = 0.46$, $\bar{\pi}_1 = 0.54$, and $q = 0.6$.

Specifically, the market maker's prior belief is $\hat{\pi}_1 \in \Pi_1$. That is, the market maker is not ambiguous about the asset payoff and, hence, not ambiguity-averse.¹² Note that the sequence $\{\hat{\pi}_t\}$ of the market maker's belief is also obtained by Bayes updating with public history; therefore, it must be that the market maker's belief is contained in the set of public beliefs, i.e., $\hat{\pi}_t \in \Pi_t$ in equilibrium.

¹² Another way to consider an ambiguity-neutral market maker is to include a second-order distribution F_t over the set of beliefs, and the market maker uses the average belief $\hat{\pi}_t = \int \pi dF_t$ to form expectations. However, this formulation will lead to additional analysis of updating the second-order beliefs. The variant model

In the variant model, informed traders still perceive ambiguity and they are averse to it. Their beliefs are around the market maker's belief. Proposition 2.5 states an informed trader's optimal trading strategy, where her trading probabilities depend on two measures of ambiguity defined as follows¹³:

$$\underline{\delta}_t := \frac{\hat{\pi}_t/(1 - \hat{\pi}_t)}{\pi_t/(1 - \pi_t)} \text{ and } \bar{\delta}_t := \frac{\bar{\pi}_t/(1 - \bar{\pi}_t)}{\hat{\pi}_t/(1 - \hat{\pi}_t)}.$$

Proposition 2.5. *In any equilibrium of the variant model, given a history h_t , the market maker's belief $\hat{\pi}_t$, and the set Π_t of public beliefs, the following holds:*

- (i) *An informed trader buys (resp., sells) surely if and only if $\underline{\delta}_t \leq \theta_*(q)$ (resp., $\bar{\delta}_t \leq \theta_*(q)$).*
- (ii) *There exists a unique $p_t^b \in (0, 1)$ (resp., $p_t^s \in (0, 1)$) such that an informed trader buys (resp., sells) with probability p_t^b (resp., p_t^s) if and only if $\theta_*(q) < \underline{\delta}_t < \theta^*(q)$ (resp., $\theta_*(q) < \bar{\delta}_t < \theta^*(q)$).*
- (iii) *An informed trader stops to buy (resp., sell) if and only if $\underline{\delta}_t \geq \theta^*(q)$ (resp., $\bar{\delta}_t \geq \theta^*(q)$).*

Note that the optimal trading strategy in the variant model exhibits a pattern similar to that of the baseline model: trading probabilities decrease as ambiguity increases. Consequently, trading can still become uninformative even when the market maker is ambiguity-neutral. The underlying intuition remains consistent with the baseline model: when ambiguity is sufficiently high, the market maker would incur losses from trading with noise traders if they were to set the ask (bid) price at or below (above) the reservation price of informed traders in order to encourage them to trade.¹⁴

3. Extensions of the baseline model

3.1. A general model

This subsection presents a general model. The model has the same timeline and players as the baseline model. However, we consider the following generalizations: 1) asset payoffs and signals can be drawn from arbitrary finite sets; 2) private signals can have ambiguous distributions; 3) informed traders can be risk-averse.

To be more specific, let $V \subset \mathbb{R}_+$ be a finite (but non-singleton) set of possible values of the asset. Let $\Delta(V)$ denote all the probability measures on the set V . In each period t , agents can have a set Π_t of probability measures, which is a closed and convex subset in $\Delta(V)$, to describe their ambiguity in the public beliefs.

The general model extends the signal space and allows ambiguity in private signals. Let $S \subset \mathbb{R}$ be a finite (but non-singleton) set of possible values of private signals and $p(\cdot|\cdot) : S \times V \rightarrow [0, 1]$ be the conditional probability of signal; that is, $p(s|v) = \Pr(s_t = s|v)$ is the probability of signal $s_t = s$ conditional on that the true value is v . Signals can also be ambiguous in the sense that informed agents consider a set of conditional probabilities $\mathcal{P}$, which is closed and convex. The set $\mathcal{P}$ is assumed to be constant over time.¹⁵ We impose that every conditional probability of private signals considered by agents satisfies the monotone likelihood ratio property (MLRP), which implies that $\Pr(\cdot|s')$ first-order stochastically dominates (FOSD) $\Pr(\cdot|s)$ if $s' > s$.

Assumption 1. Any conditional probability distribution $p \in \mathcal{P}$ satisfies MLRP, i.e.,

$$\frac{p(s'|v')}{p(s'|v)} \geq \frac{p(s|v')}{p(s|v)},$$

whenever $s' \geq s$ and $v' \geq v$, with strict inequality if $s' > s$, or $v' > v$, or both.

Moreover, the general model allows informed traders to be risk-averse. Specifically, their objective is to maximize the min-expected utility as follows:

$$\max_{\sigma_t \in \Sigma_t} \min_{p \in \mathcal{P}, \pi \in \Pi_t(h_t)} E^{p \times \pi} [\zeta_t u(v - a_t) + \iota_t u(b_t - v) | h_t, a_t, b_t, s_t], \quad (3.1)$$

where u is a von Neumann-Morgenstern (vNM) utility function, and we impose the following assumption.

in this subsection suffices to make the point that the market maker's ambiguity attitude does not affect the results of the baseline model substantially without the complication of updating second-order beliefs.

¹³ The proof is omitted since it is similar to the proof of Proposition 2.1. The presence of two separate ambiguity measures, compared to the single measure δ_t in the baseline model, stems from the heterogeneous beliefs between the ambiguity-neutral market maker and the ambiguity-averse informed traders. With a common prior set, all agents share a single level of ambiguity δ_t . However, with heterogeneous prior sets, the market maker and informed traders use different belief sets to evaluate a transaction, which necessitates distinct measures of ambiguity for buy and sell orders.

¹⁴ In fact, the results would be similar if we assume that only the market maker is ambiguity-averse but traders are ambiguity-neutral. The analysis and logic are also similar. Therefore, our results regarding the possibility of mixed strategies and no-trade hold generally when either or both sides of the market are ambiguity-averse.

¹⁵ This assumption can be justified by that the ambiguous signals are conditionally independent in the sense that they are *epistemically independent* (Couso et al., 1999) conditional on value $v \in V$ of the asset. Loosely speaking, knowing the past signals will not change the perceived set of conditional probabilities of future signals. Moreover, the joint distribution of value v and signal s at period t is $\mathcal{D}_t = \{p \times \pi : \pi \in \Pi_t, p \in \mathcal{P}\}$.

Assumption 2. u is continuous, concave, and strictly increasing; and $u(0) = 0$.

In the general model, pricing rules, strategies, and beliefs are adapted to the extended payoff and signal spaces. The solution concept is also similarly defined as the baseline model: pricing rules satisfy zero profit conditions and trading decisions are optimal given the belief system, whereas the concept of belief consistency is extended in the general model. Here, it is required that a belief system Π is consistent w.r.t. Π_1 , $\mathcal{P}$, and (x, σ) ; that is, for any $t > 1$ and history h_t , for any $\pi_t \in \Pi_t(h_t)$, there exists some $\pi_1 \in \Pi_1$ and $p \in \mathcal{P}$ such that π_t is updated by Bayes rule given π_1 , p , and (x, σ) wherever possible.

The first theorem asserts the existence of the equilibrium in the general model. In fact, the zero profit conditions implicitly represent a fixed-point characterization of the equilibrium. Thus, the proof can be done by a fixed-point argument.¹⁶

Theorem 3.1. *There exists an equilibrium (x, σ, Π) for the trading game.*

The next theorem characterizes the trading strategies in equilibrium. The signal set S is finite and non-singleton, so there exists a minimum and a maximum element, which denoted by $\underline{s} := \min S$ and $\bar{s} := \max S$, respectively. The equilibrium trading strategies are *monotonic*; that is, it can be described as such: there are two cutoff signals, $s_* \geq \underline{s}$ and $s^* \leq \bar{s}$, with $s_* < s^*$, and the active informed traders will buy the asset if her private signal is at least s^* and sell the asset if her private signal is no more than s_* , where the determination of s^* and s_* depends on the set of common beliefs. In the case that $s_* = \underline{s}$ and $s^* = \bar{s}$, trading probabilities can be always zero: informed traders do not trade, and social learning stops.

Theorem 3.2. *In any equilibrium (x, σ, Π) , given any history h_t on equilibrium path and prices $(a_t, b_t) = x_t(h_t)$, it must be that there exists some $s_{t*}, s_t^* \in S$ such that $\underline{s} \leq s_{t*} < s_t^* \leq \bar{s}$ and*

$$\zeta_t(h_t, a_t, b_t, s) \begin{cases} = 0 & \forall s < s_{t*} \\ \geq 0 & s = s_{t*} \\ = 1 & \forall s > s_t^* \end{cases}, \quad \iota_t(h_t, a_t, b_t, s) \begin{cases} = 1 & \forall s < s_{t*} \\ \geq 0 & s = s_{t*} \\ = 0 & \forall s > s_t^* \end{cases}.$$

The general model allows further explore the implications of the effect ambiguity in a more general information structure than the baseline model. In the following, Section 3.2 shows that ambiguity in signals can lead to information aggregation failure in the long run with ambiguous signal accuracy, even though traders' actions are informative at the beginning. Section 3.3 considers that private signals are asymmetric and shows that no-trade can be informative.

3.2. Ambiguous signal accuracy

In an equilibrium of the baseline model, the level of ambiguity is constant over time on an equilibrium path. To overcome it, we consider a *model with ambiguous signal accuracy* that extends the baseline model by introducing ambiguity into private signals. Specifically, instead of a commonly known, unique signal accuracy q , agents are uncertain about signal accuracy, and they consider that

$$\Pr(s_t = 1 | v = 1) = \Pr(s_t = 0 | v = 0) = \tilde{q},$$

which can be any value in an interval $Q := [q, \tilde{q}]$ with $1/2 < q < \tilde{q} < 1$.

Beliefs are updated by a full Bayesian rule, and it is done in a prior-by-prior and likelihood-by-likelihood manner in the model with ambiguous signal accuracy. For a prior belief π_t and a signal accuracy $\tilde{q}$, its posterior belief $\pi_t^{s_t, \tilde{q}}$ given a realized signal s_t is obtained by Bayes rule, i.e.,

$$\pi_t^{s_t, \tilde{q}} = \begin{cases} \frac{\pi_t \tilde{q}}{\pi_t \tilde{q} + (1 - \pi_t)(1 - \tilde{q})} & \text{if } s_t = 1, \\ \frac{\pi_t (1 - \tilde{q})}{\pi_t (1 - \tilde{q}) + (1 - \pi_t) \tilde{q}} & \text{if } s_t = 0. \end{cases}$$

For a set Π_t of prior beliefs and a set Q of signal accuracies, the set of posterior beliefs given a realized signal s_t is obtained by updating each $\pi_t \in \Pi_t$ with each signal accuracy $\tilde{q} \in Q$, i.e., $\Pi_t^{s_t} = \left\{ \pi_t^{s_t, \tilde{q}} \mid \pi_t \in \Pi_t, \tilde{q} \in Q \right\}$.

Similar to the baseline model, the market maker posts prices satisfying the zero profit conditions. However, ambiguity aversion makes the market maker pick the worst posterior belief against the trading direction. That is, given a history h_t and the corresponding set Π_t of public beliefs, ask and bid prices are set as follows:

$$a_t = \max_{\pi_t \in \Pi_t, \tilde{q} \in Q} E^{\pi_t, \tilde{q}} [v | h_t, z_t = 1],$$

$$b_t = \min_{\pi_t \in \Pi_t, \tilde{q} \in Q} E^{\pi_t, \tilde{q}} [v | h_t, z_t = -1].$$

¹⁶ The proof of the theorems in this subsection can be found in my thesis (Chu, 2018) or available upon request.

An ambiguity-averse informed trader also trades based on the worst posterior belief against her trading direction. Therefore, an informed trader decides to buy (with some probability) if $\min \Pi_t^1 = \pi_t^{1,q} \geq a_t$ and to sell if $\max \Pi_t^0 = \pi_t^{0,q} \leq b_t$.

To understand the effect of ambiguous signal accuracy, we can consider a 3-color urn: there are 90 balls in an urn with 30 being red and the remaining 60 balls being either blue or green.¹⁷ Let f_{BG} denote the lottery paying 1 if a blue or green ball is drawn, zero otherwise. Without additional information, this lottery is risky but not ambiguous; the probability of winning is $2/3$. Instead, suppose that someone else first draws a ball and reveals that the ball is not green. This information serves as an ambiguous signal.¹⁸ Then, the lottery f_{BG} becomes ambiguous, the probability of winning ranging from 0 to $2/3$, conditional on the signal.

Back to our model, the observed order z_t is a noisy observation of an informed trader's private signal from the public perspective. Therefore, if a private signal has ambiguous accuracy, the observed order is also ambiguous, i.e., it does not have a unique conditional distribution. Hence, even though there is no ambiguity in the public belief initially, observing an ambiguous trading order makes posterior belief ambiguous. With a sequence of signals with ambiguous accuracy, ambiguity in public beliefs grows over time.

Proposition 3.1. *Consider a model with ambiguous signal accuracy, in which the initial set Π_1 of public beliefs satisfies that $\delta_1 < \theta^*(q)$. Then $\{\delta_t\}_{t=1}^\infty$ is an increasing sequence on an equilibrium path, and it converges to $\theta^*(q)$ as $t \rightarrow \infty$ almost surely.*

Similar to the baseline model, the sufficient and necessary conditions for an informed trader to trade with positive probability is $\delta_t < \theta^*(q)$ in the model with ambiguous signal accuracy.¹⁹ In this case, the order z_t is a noisy observation of ambiguous private signals, making itself an ambiguous signal. Conversely, if $\delta_t \geq \theta^*(q)$, informed traders do not trade, and the order z_t is uninformative about private signals. Thus, the market enters a state where social learning stops, and the set of public beliefs remains constant over time.

When $\delta_t < \theta^*(q)$, the order z_t may contain ambiguous information about private signal accuracy. Hence, public beliefs suffer from dilution: the ambiguity of private signal accuracy is injected into the set Π_{t+1} of public beliefs through updating with z_t . Consequently, $\delta_{t+1} \geq \delta_t$ with strict inequality if $z_t \neq 0$.²⁰ Inductively, the level of ambiguity is an increasing sequence.

Particularly, when the ambiguity is of a moderate level, i.e., $\theta_*(q) < \delta_t < \theta^*(q)$, an informed trader mixes between trading or not. The higher the level of ambiguity, the smaller probability an informed trader trades with. The small trading probability chosen by the informed trader makes the observed order slightly contaminated by the ambiguity in the private signal. Hence, it restricts the incremental of ambiguity in the public beliefs, and the level of ambiguity never surpasses $\theta^*(q)$.

However, in the long run, the increasing sequence of the level of ambiguity converges to $\theta^*(q)$ almost surely. As the level of ambiguity increases over time, an informed trader's trading probability decreases and converges to zero in the long run; that is, the market gets arbitrarily close to a noninformative trading regime. The following corollary follows Proposition 3.1 immediately.

Corollary 3.1. *Consider a model with ambiguous signal accuracy with an initial set Π_1 of public beliefs satisfying $\delta_1 < \theta^*(q)$. It converges to a noninformative trading regime almost surely on an equilibrium path.*

Fig. 3.1 illustrates the dynamics of prices and beliefs in the model with ambiguous signal accuracy. Even with small ambiguity ($q = 0.6$ and $\bar{q} = 0.61$) in the signal accuracy, we can see that after several hundred of periods, social learning is almost ceased, and the beliefs and prices becomes almost flat.

3.3. Asymmetric signals

If no-trade can result from an informed trader's rational decision, it can also convey information about an informed trader's private signal. However, it is not the case in the baseline model where signals are assumed to be symmetric. To make no-trade possibly informative, we breakdown the symmetry assumption in the baseline model and consider a *model with asymmetric signals*, in which the assumption (2.3) is replaced by

$$q_1 = \Pr(s = 1 | v = 1) > \Pr(s = 0 | v = 0) = q_0,$$

with other assumptions in the baseline model unchanged.

In the baseline model, an informed trader buys upon a good signal and sells upon a bad signal with the same trading probability p . It is a consequence of the symmetry assumption on signal accuracy. By introducing asymmetry, an informed trader's *buy probability*, $\hat{c}_t$ and *sell probability*, $\hat{i}_t$, can be different.

Proposition 3.2. *In an equilibrium of the model with asymmetric signals where $q_1 > q_0$, $0 \leq \hat{i}_t < \hat{c}_t \leq 1$ if and only if*

$$\theta_*(q_0) < \delta_t < \theta^*(q_1). \quad (3.2)$$

¹⁷ This example can be also found in Ozdenoren and Peck (2008).

¹⁸ The probability $\Pr(RB|BG)$ of the ball being not green, i.e., red or blue, conditional on the ball is blue or green ranges from 0 to 1.

¹⁹ There are two cases when $\delta_t < \theta^*(q)$. If $\delta_t \leq \theta_*(q)$, an informed trader would trade surely. If $\theta_*(q) < \delta_t < \theta^*(q)$, an informed trader would mix between trading and non-trading.

²⁰ In the case that $z_t = 0$, the signal is uninformative to the market maker.

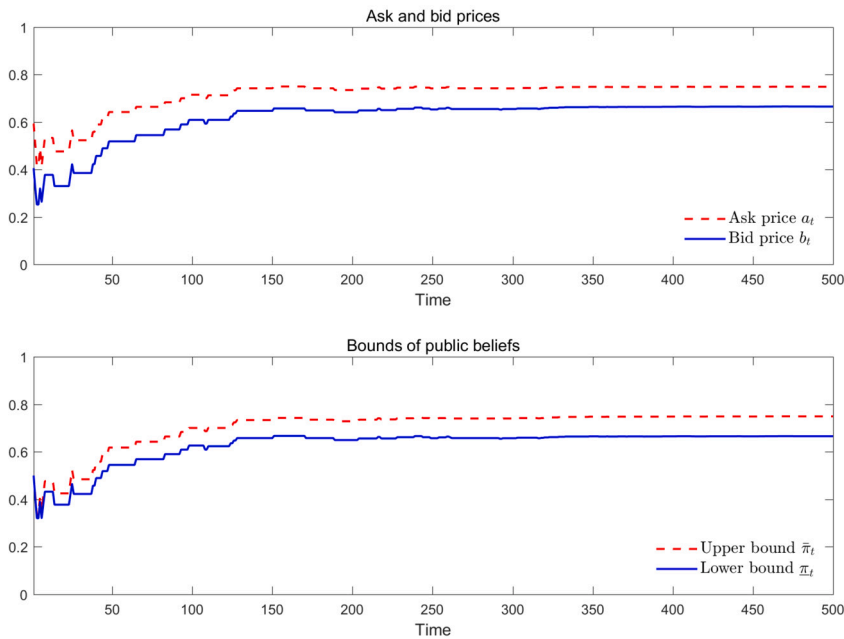


Fig. 3.1. The model with ambiguous signal accuracy ($v = 1$, $\Pi_1 = \{0.5\}$, $\underline{q} = 0.6$, $\bar{q} = 0.61$, $\eta = 0.8$).

Moreover, if observing order $z_t = 0$, public evaluation of the asset value decreases in the sense that $\Pi_{t+1} \neq \Pi_t$ with $\underline{\pi}_{t+1} < \underline{\pi}_t$ and $\bar{\pi}_{t+1} < \bar{\pi}_t$.

The proof of Proposition 3.2 follows a similar approach as that of Proposition 2.1, and hence it is omitted. Instead, we discuss several cases in terms of the ambiguity level, δ_t , with a fixed history h_t and a set Π_t of public beliefs given. Before discussing the case when inequality (3.2) holds, let us first see what if it does not. If $\delta_t \geq \theta^*(q_1)$, the level of ambiguity is so high that both $\underline{\pi}_t^{(1)} \leq \bar{\pi}_t$ and $\bar{\pi}_t^{(0)} \geq \underline{\pi}_t$ hold. Similar to the baseline model, the market maker posts an ask price $\bar{\pi}_t$ and a bid price $\underline{\pi}_t$, and an informed trader finds it optimal not to trade. It makes both buy probability and sell probability equal to zero.

On the other hand, if the level of ambiguity is sufficiently low, i.e., $\delta_t \leq \theta_*(q_0)$, an informed trader trades with certainty as in the baseline model; that is, her buy probability and sell probability are both equal to one. If buy probability and sell probability are equal, the likelihood of observing $z_t = 0$ is also equal conditional on different true asset values. Then, $z_t = 0$ is not informative about the true asset value.

However, ambiguity-averse informed traders may mix between trading or not when the level of ambiguity is moderate, which can make no-trade informative when it is combined with asymmetry in signal accuracy. Consider different situations when inequality (3.2) holds. If $\theta_*(q_0) < \delta_t \leq \theta_*(q_1)$, an informed trader with a good signal views that the level of ambiguity is small, while an informed trader with a bad signal considers that the level of ambiguity is moderate. It follows that an informed trader buys the asset with certainty upon a good signal while selling the asset with a probability less than one upon a bad signal.

If $\theta^*(q_0) \leq \delta_t < \theta^*(q_1)$, it is the case that an informed trader with a good signal thinks that the level of ambiguity is moderate while an informed trader with a bad signal feels that the level of ambiguity is large. It follows that an informed trader buys the asset with a positive probability upon a good signal but never sells the asset upon a bad signal. Put it differently, it is likely that an informed trader has seen a bad signal if there is no trade. Moreover, a sell order, $z_t = -1$, is not informative about the asset value since it must come from a noise trader.

If $\theta^*(q_0) \leq \theta_*(q_1)$, it is possible that $\theta^*(q_0) \leq \delta_t \leq \theta_*(q_1)$. There is an overlap of the two situations mentioned above, where an informed trader buys the asset upon a good signal and never sells the asset. If $\theta_*(q_1) < \theta^*(q_0)$, it can be the case that $\theta_*(q_1) < \delta_t < \theta^*(q_0)$. Then an informed trader mixes between trading or not whenever seeing a good or bad signal. However, since a bad signal is less accurate than a good signal, the informed trader is more conservative in placing a sell order upon a bad signal than placing a buy order upon a good signal. Thus, an informed trader's sell probability is strictly less than buy probability in this case.

To sum up, no-trade is more likely to result from a bad signal than a good signal when inequality (3.2) holds. Therefore, if no-trade occurs, the valuation of the asset will decrease in the market: no news is bad news, given that a bad signal is less accurate than a good one.

3.4. Continuous action spaces

Besides the price critique, Chari and Kehoe (2004) also point out a continuous investment critique (Lee, 1993) on the possibility of information aggregation failure. That is, the informational cascade disappears if agents' action spaces are continuous. Moreover, Decamps and Lovo (2006) show that, with risk-averse traders, a “non-informative trade” can occur when prices are endogenous. In

their model, discreteness in action spaces is a necessary condition. However, in our model, the non-informative trading regime with ambiguity-averse agents is robustly attributed to the non-differentiability of the multiple prior preferences, and it can occur even when traders have continuous action spaces.

Let us consider a *model with continuous action spaces*; specifically, the baseline model is extended by allowing the informed traders to buy or sell the risky asset at any amount up to one unit. An order z_t can take any value in the interval $[-1, 1]$. A positive number indicates a buy order, and a negative number is a sell order. Moreover, a noise trader's order is randomly drawn from a continuous distribution with support $[-1, 1]$. This assumption can be justified by the fact that noise traders have different demands for the asset. Moreover, it also guarantees that an informed trader is indistinguishable from a noise trader. For simplicity, assume that a noise trader's order is drawn from a uniform distribution. The market maker, besides posting prices, is also allowed to reject some trades. We maintain other assumptions in the baseline model.

Instead of solving all the possible equilibria, we aim to find an equilibrium where a non-informative trading may exist. For this purpose, we construct a set of trading strategies and pricing rules that can be *supported in equilibrium*. That is, there exists a belief system, along with which the trading strategies and pricing rules constitute an equilibrium.

For an informed trader, we conjecture that their optimal trading decisions have a similar pattern as those in the baseline model. If the level of ambiguity is low, she trades with certainty; if the level of ambiguity is moderate, she mixes between trading and non-trading; if the level of ambiguity is high, she does not trade at all. Moreover, it is reasonable to expect the prices posted by the market maker to make an informed trader trade with random order size to mimic a noise trader. If an informed trader always places an order with a fixed size, the market maker can distinguish them from noise traders.

To further understand how the market maker sets prices, let us focus on the intuition of setting an ask price in the following. Consider that the level of ambiguity is sufficiently low, and suppose that an informed trader must place a buy order upon a good signal. If the market maker accepts trade of any size, ask price must be $\max_{\pi_t \in \Pi_t} E^{\pi_t} [v|h_t, z_t > 0]$ for each unit of the asset to conform with the zero profit conditions. However, the low level of ambiguity implies that an informed trader's worst belief $\underline{\pi}_t^{(1)}$ can be strictly higher than $\max_{\pi_t \in \Pi_t} E^{\pi_t} [v|h_t, z_t > 0]$. Thus, an informed trader will always place an order of size one to maximize her expected profit. If it is the case, such an adverse selection allows the market maker to distinguish an informed trader from a noise trader.

Nevertheless, the market maker can reject orders of size above a cutoff $z^* \in [0, 1]$ to make it possible that ask price equal to an informed trader's worst belief upon a good signal, and thus an informed trader trade with random size. Restricting the order size forces the informed traders to place an order of size below z^* . Moreover, by lowering the cutoff z^* of order size, it increases the likelihood of that an acceptable buy order is placed by an informed trader given that an informed trader trades with certainty. Thus, the market maker can increase the ask price (for orders of acceptable size) to match the worst belief of an informed trader upon a good signal.

When the level of ambiguity is moderate, informed traders trade with some probability $p \in (0, 1)$. Suppose that informed traders still trade with certainty given a moderate level of ambiguity. Then, to meet zero profit conditions, the ask price posted by the market maker would be strictly higher than an informed trader's worst belief upon seeing a good signal, i.e., $\underline{\pi}_t^{(1)}$. However, by choosing a probability $p < 1$, it reduces the informativeness of the signal $\mathbf{1}_{\{z_t > 0\}}$ to the market maker and makes the break-even ask price equal to $\underline{\pi}_t^{(1)}$ exactly. However, when ambiguity is sufficiently high so that $\bar{\pi}_t \geq \underline{\pi}_t^{(1)}$, there is no way to post an ask price to make an informed trader to trade while the market maker does not incur a loss. As the baseline model, the market maker simply posts an ask price equal to $\bar{\pi}_t$ to guarantee zero profit from noise traders, and no informed traders want to trade.

We verify that the above construction of trading strategies and pricing rule can be supported in equilibrium, and summarize this result in Proposition 3.3. It also verifies that there exists an equilibrium where there can be a non-informative trading regime if ambiguity is sufficiently high.

Proposition 3.3. *Consider a model with a continuous action space with an initial set Π_1 of public beliefs.*

- (i) *If $1 < \delta_1 < \theta^*(q)$, then the following can be supported in an equilibrium: (i.a) There exists a $z^* \in (0, 1]$ such that the market maker rejects all trades of size above z^* , and the ask and bid prices are $\underline{\pi}_t^{(1)}$ and $\bar{\pi}_t^{(0)}$, respectively, in each period $t \geq 1$. (i.b) There exists a probability $p \in (0, 1]$ such that an informed trader trades with probability p , and her trading size is drawn randomly from a uniform distribution with support $[0, z^*]$.*
- (ii) *If $\delta_1 \geq \theta^*(q)$, then the following can be supported in an equilibrium: (ii.a) The market maker accepts all trades, and ask and bid prices are $\bar{\pi}_t$ and $\underline{\pi}_t$, respectively. (ii.b) An informed trader never trades.*

Therefore, the non-informative trading regime, which leads to failure of information aggregation, emerges when the ambiguity is high, even though traders' action spaces are continuous. Such an information aggregation failure that arises from agents' multiple prior preferences is robust to endogenous prices and continuous action spaces. As aforementioned, a no-trade region exists for traders with multiple prior preferences due to the non-differentiability in their indifference curves. When ambiguity is sufficiently high, the market maker wants a sufficiently large ask–bid spread to elicit informed traders to trade. However, its pricing cannot be competitive in the market.

4. Concluding remarks

In this paper, we examine how ambiguity aversion affects sequential trading. Ambiguity-averse traders trade less with private information when they face ambiguous information. They may avoid holding any uncertain asset if the ambiguity is too high. The main contribution of the paper is to show that ambiguity aversion may cause information aggregation failure even with endogenous prices and continuous actions spaces. In our model, the key reason for these results is attributed to the non-smoothness of the multiple prior preferences. Therefore, we conjecture that the result can be extended to a broader class of ambiguity-averse preferences, for example, Choquet expected utility model (Schmeidler, 1989) and α -maxmin model (Ghirardato et al., 2004).²¹

Our results hold for other model extensions, such as ambiguous signal accuracy, general finite signal and action spaces, and continuous action spaces. We also find interesting results in these extensions. For example, when signals are ambiguous, trading activities make public beliefs more ambiguous over time. This depends on a key property that full Bayesian updating widens the posterior beliefs.²² Moreover, our model shows how no-trade can inform the market when ambiguity-averse traders may abstain. The type of information conveyed by no-trade depends on signal accuracy. For example, if bad signals are less accurate than good signals, informed traders with bad signals order less. Thus, no news is bad news for the market, and prices can drop when trading is inactive.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Proofs

Proof of Proposition 2.1. We only prove the results regarding $\hat{\zeta}_t$, and the proof of results regarding $\hat{\eta}_t$ is similar. We first show the if sides of (i)–(iii).

(i) Let the inequality $\delta_t \leq \theta_*(q)$ hold. And suppose, for contradiction, $\hat{\zeta}_t < 1$. Then, there are two cases.

Case (i.a): $\hat{\zeta}_t \in (0, 1)$. In this case, by zero profit condition (2.1),

$$a_t = \frac{\bar{\pi}_t(q + \gamma/\hat{\zeta}_t)}{\bar{\pi}_t(q + \gamma/\hat{\zeta}_t) + (1 - \bar{\pi}_t)(1 - q + \gamma/\hat{\zeta}_t)}.$$

However, the fact that an informed trader mixes buying and not buying upon a good signal implies that $a_t = \underline{\pi}_t^{(1)}$, i.e.,

$$a_t = \frac{\underline{\pi}_t q}{\underline{\pi}_t q + (1 - \underline{\pi}_t)(1 - q)}.$$

Therefore, we obtain that

$$\delta_t = \frac{q}{1 - q} \frac{1 - q + \gamma/\hat{\zeta}_t}{q + \gamma/\hat{\zeta}_t}. \quad (\text{A.1})$$

Note that $\hat{\zeta}_t < 1$ implies that

$$\frac{1 - q + \gamma/\hat{\zeta}_t}{q + \gamma/\hat{\zeta}_t} > \frac{1 - q + \gamma}{q + \gamma}. \quad (\text{A.2})$$

Then, we obtain a contradiction that $\delta_t > \theta_*(q)$ from inequalities (A.1) and (A.2).

Case (i.b) $\hat{\zeta}_t = 0$. In this case, it must be that $a_t = \bar{\pi}_t$ since all buy orders come from noise traders. Moreover, by Lemma 2.1, $\hat{\zeta}_t = 0$ implies that $\underline{\pi}_t^{(1)} \leq a_t$. Hence, it follows that

$$\underline{\pi}_t^{(1)} \leq \bar{\pi}_t \iff \delta_t \geq \theta^*(q).$$

Since $\theta^*(q) > \theta_*(q)$, we again obtain a contradiction that $\delta_t > \theta_*(q)$.

(ii) Let inequality (2.7) hold.

Suppose, for contradiction, $\hat{\zeta}_t = 1$. In this case, we can explicitly compute ask price, which equals to a_t^* as defined in (2.5). Then, for an informed trader optimally choosing $\zeta_t = 1$, by Lemma 2.1, it follows that $\underline{\pi}_t^{(1)} \geq a_t^*$. It further implies that $\delta_t \leq \theta_*(q)$, which contradicts with inequality (2.7).

Suppose, for contradiction, $\hat{\zeta}_t = 0$. Following a similar argument as Case (i.b), it implies that $\delta_t \geq \theta^*(q)$, which contradicts with inequality (2.7).

²¹ In a more general class of ambiguity-averse preferences, the variational preferences, Maccheroni et al. (2006) show that the preferences are not smooth if their ambiguity indices are not strictly convex.

²² Our results in the model of ambiguous signal accuracy may not be true if belief updating follows a maximum likelihood rule (Gilboa and Schmeidler, 1993). However, for other alternative approaches that are not as extreme as the maximum likelihood rule, for example, belief updating via a likelihood test (Epstein and Schneider, 2007) or a relative maximum likelihood rule (Cheng, 2022), our results are still valid.

Hence, we conclude that it must be that $\zeta_t \in (0, 1)$. Moreover, following the argument in Case (i.b), equation (A.1) must hold. Its right-hand side strictly decreases with $\hat{\zeta}_t$ with

$$\lim_{\zeta_t \rightarrow 0} \frac{q}{1-q} \frac{1-q+\gamma/\hat{\zeta}_t}{q+\gamma/\hat{\zeta}_t} = \frac{q}{1-q} > \delta_t,$$

$$\lim_{\zeta_t \rightarrow 1} \frac{q}{1-q} \frac{1-q+\gamma/\hat{\zeta}_t}{q+\gamma/\hat{\zeta}_t} = \frac{q}{1-q} \frac{1-q+\gamma}{q+\gamma} < \delta_t.$$

Thus, there must exist a unique $p_t \in (0, 1)$ that makes equality (A.1) hold when $\hat{\zeta}_t = p_t$. Moreover, Corollary 2.1 directly follows the fact that right-hand side of equation (A.1) strictly decreases with $\hat{\zeta}_t$.

(iii) Let the inequality $\delta_t > \theta^*(q)$ hold. Suppose, for contradiction, $\hat{\zeta}_t > 0$.

Case (iii.a): $\hat{\zeta}_t \in (0, 1)$. Following a similar argument as Case (i.a), it implies that

$$\delta_t = \frac{q}{1-q} \frac{1-q+\gamma/\zeta_t}{q+\gamma/\zeta_t} < \theta^*(q),$$

a contradiction.

Case (iii.b): $\hat{\zeta}_t = 1$. Following a similar argument as Case (ii), it also implies that $\delta_t < \theta_*(q)$, a contradiction.

For only if sides of (i)–(iii), we simply reverse the above arguments. $\square$

Proof of Proposition 2.2. Let $\Pi'_t = [\underline{\pi}'_t, \bar{\pi}'_t]$ and $\Pi_t = [\underline{\pi}_t, \bar{\pi}_t]$ be given, and Π'_t is more ambiguous than Π_t . Π'_t is more ambiguous than Π_t implies that

$$\delta'_t := \frac{\bar{\pi}'_t(1-\underline{\pi}'_t)}{\underline{\pi}'_t(1-\bar{\pi}'_t)} > \delta_t := \frac{\bar{\pi}_t(1-\underline{\pi}_t)}{\underline{\pi}_t(1-\bar{\pi}_t)}.$$

Moreover, let p'_t and p_t be the corresponding trading probability given Π'_t and Π_t .

First, an equilibrium ask price, given $\Pi_t = [\underline{\pi}_t, \bar{\pi}_t]$ and p_t , can be written explicitly by

$$a_t = \frac{\bar{\pi}_t(\gamma + qp_t)}{\gamma + [\bar{\pi}_t q + (1 - \bar{\pi}_t)(1 - q)] p_t},$$

which strictly increases with $\bar{\pi}_t$ and p_t . Moreover, by Corollary 2.1, $\delta'_t > \delta_t$ implies that $p'_t \leq p_t$. Hence,

$$a'_t = \frac{\bar{\pi}'_t(\gamma + qp'_t)}{\gamma + [\bar{\pi}'_t q + (1 - \bar{\pi}'_t)(1 - q)] p'_t}$$

$$> \frac{\bar{\pi}_t(\gamma + qp_t)}{\gamma + [\bar{\pi}_t q + (1 - \bar{\pi}_t)(1 - q)] p_t} = a_t,$$

where the inequality holds due to that $\bar{\pi}'_t > \bar{\pi}_t$ and $p'_t \leq p_t$. It can be similarly shown that $b'_t < b_t$, and our desired result follows immediately. $\square$

Proof of Proposition 2.3. Fix an equilibrium and a history h_t on equilibrium path. For an arbitrary $z_t \in Z$, use l_t as short for $l_t(z_t)$.

$$\begin{aligned} \bar{\pi}_{t+1} &= \max \Pi_{t+1}(h_t, a_t, b_t, z_t) \\ &= \max_{\pi_t \in \Pi_t} \frac{\pi_t \Pr(z_t | h_t, v=1)}{\pi_t \Pr(z_t | h_t, v=1) + (1 - \pi_t) \Pr(z_t | h_t, v=0)} \\ &= \frac{\bar{\pi}_t \Pr(z_t | h_t, v=1)}{\bar{\pi}_t \Pr(z_t | h_t, v=1) + (1 - \bar{\pi}_t) \Pr(z_t | h_t, v=0)}, \end{aligned}$$

therefore,

$$\frac{\bar{\pi}_{t+1}}{1 - \bar{\pi}_{t+1}} = \frac{\bar{\pi}_t}{1 - \bar{\pi}_t} l_t.$$

Similarly,

$$\frac{\underline{\pi}_{t+1}}{1 - \underline{\pi}_{t+1}} = \frac{\underline{\pi}_t}{1 - \underline{\pi}_t} l_t.$$

Hence,

$$\frac{\bar{\pi}_{t+1}}{1 - \bar{\pi}_{t+1}} \frac{1 - \underline{\pi}_{t+1}}{\underline{\pi}_{t+1}} = \frac{\bar{\pi}_t}{1 - \bar{\pi}_t} \frac{1 - \underline{\pi}_t}{\underline{\pi}_t},$$

and rest of the proof is done by mathematical induction. $\square$

Proof of Proposition 3.1. Consider an equilibrium in the model with ambiguous signal accuracy. First, we show that δ_t is increasing over time. Further consider a period t in a history h_t on equilibrium path. As a similar argument in the baseline model, given that $\delta_t < \underline{q}/(1 - \underline{q})$, an informed trader's strategy must be that $\hat{\zeta}_t = \hat{\iota}_t = p_t$ for some $p_t \in (0, 1]$. Then, consider an extended likelihood function $\hat{\iota}_t(z_t; \bar{q})$ as follows

$$\hat{\iota}_t(z_t; \bar{q}) = \begin{cases} \frac{\bar{q}p_t + \gamma}{(1 - \bar{q})p_t + \gamma} & \text{if } z_t = 1, \\ 1 & \text{if } z_t = 0, \\ \frac{(1 - \bar{q})p_t + \gamma}{\bar{q}p_t + \gamma} & \text{if } z_t = -1. \end{cases}$$

By updating beliefs in a prior-by-prior and likelihood-by-likelihood manner,

$$\begin{aligned} \frac{\bar{\pi}_{t+1}}{1 - \bar{\pi}_{t+1}} &= \frac{\bar{\pi}_t}{1 - \bar{\pi}_t} \max_{\bar{q} \in Q} \hat{\iota}_t(z_t; \bar{q}) = \frac{\bar{\pi}_t}{1 - \bar{\pi}_t} \hat{\iota}_t(z_t; \bar{q}), \\ \frac{\underline{\pi}_{t+1}}{1 - \underline{\pi}_{t+1}} &= \frac{\underline{\pi}_t}{1 - \underline{\pi}_t} \min_{\bar{q} \in Q} \hat{\iota}_t(z_t; \bar{q}) = \frac{\underline{\pi}_t}{1 - \underline{\pi}_t} \hat{\iota}_t(z_t; \underline{q}). \end{aligned}$$

Therefore,

$$\begin{aligned} \delta_{t+1} &= \delta_t \frac{\hat{\iota}_t(z_t; \bar{q})}{\hat{\iota}_t(z_t; \underline{q})} \\ &= \delta_t \frac{\bar{q}p_t + \gamma}{(1 - \bar{q})p_t + \gamma} \frac{(1 - \underline{q})p_t + \gamma}{\underline{q}p_t + \gamma} > \delta_t \text{ if } z_t \neq 0, \end{aligned}$$

whereas the $\delta_{t+1} = \delta_t$ if $z_t = 0$.

We next show that $\delta_t < \underline{q}/(1 - \underline{q})$ for all $t = 1, 2, \dots$, on equilibrium path by mathematical induction. It is true when $t = 1$ by our assumption, and suppose that it is true for some $t < \infty$ in a history h_t on equilibrium path. We show that $\delta_{t+1} < \underline{q}/(1 - \underline{q})$. If $z_t = 0$, then $\max_{\bar{q} \in Q} \hat{\iota}_t(z_t; \bar{q}) / \min_{\bar{q} \in Q} \hat{\iota}_t(z_t; \bar{q}) = 1$, and it follows immediately that $\delta_{t+1} = \delta_t < \underline{q}/(1 - \underline{q})$. Suppose that $z_t \neq 0$. First, note that, if $z_t \neq 0$,

$$\frac{\hat{\iota}_t(z_t; \bar{q})}{\hat{\iota}_t(z_t; \underline{q})} = \frac{\bar{q}p_t + \gamma}{(1 - \bar{q})p_t + \gamma} \frac{(1 - \underline{q})p_t + \gamma}{\underline{q}p_t + \gamma}.$$

Second, in equilibrium, given an informed trader's probability p_t of trading, it can be shown that

$$\delta_t \leq \frac{\underline{q}}{1 - \underline{q}} \frac{(1 - \bar{q})p_t + \gamma}{\bar{q}p_t + \gamma}$$

with equality hold when $\delta_t \geq \frac{\underline{q}}{1 - \underline{q}} \frac{1 - \bar{q} + \gamma}{\bar{q} + \gamma}$. By the above argument,

$$\begin{aligned} \delta_{t+1} &= \delta_t \frac{\hat{\iota}_t(z_t; \bar{q})}{\hat{\iota}_t(z_t; \underline{q})} \\ &\leq \frac{\underline{q}}{1 - \underline{q}} \frac{(1 - \bar{q})p_t + \gamma}{\bar{q}p_t + \gamma} < \frac{\underline{q}}{1 - \underline{q}}, \end{aligned}$$

where the last inequality follows that $(1 - \bar{q})p_t + \gamma / \bar{q}p_t + \gamma < 1$.

Fix some $\varepsilon > 0$, I claim that there exists some $T_\varepsilon < \infty$ such that $\delta_{T_\varepsilon} > \underline{q}/(1 - \underline{q}) - \varepsilon$ if $z_t \neq 0$ for all $t = 1, \dots, T_\varepsilon$. This is because δ_t is a strictly increasing sequence given $z_t \neq 0$ with supremum $\underline{q}/(1 - \underline{q})$. Then, we can show that for $t \geq T_\varepsilon$ large enough,

$$\begin{aligned} \Pr \left(\left| \delta_t - \frac{\underline{q}}{1 - \underline{q}} \right| > \varepsilon \right) &= \Pr \left(\# \{ z_\tau \neq 0, \tau \leq t \} < T_\varepsilon \right) \\ &= \sum_{k < T_\varepsilon} \binom{t}{k} \left(\frac{2(1 - \eta)}{3} + \eta p_t \right)^k \left(\frac{1 - \eta}{3} + \eta(1 - p_t) \right)^{t-k} \\ &\leq \sum_{k < T_\varepsilon} \binom{t}{k} / 2^t. \end{aligned}$$

Note that $\sum_{t \geq T_\varepsilon} \sum_{k < T_\varepsilon} \binom{t}{k} / 2^t < \infty$, so we have $\sum_{t=1}^{\infty} \Pr \left(\left| \delta_t - \frac{\underline{q}}{1 - \underline{q}} \right| > \varepsilon \right) < \infty$, and we can conclude the almost sure convergence. $\square$

Proof of Proposition 3.3. We simply construct a belief system consistent to the trading strategies and pricing rules specified in case (i) and (ii), respectively, and show that they support the corresponding strategies and rules in equilibrium.

Case (i): Consider a belief system consistent to the trading strategy and pricing rule specified in case (i), where beliefs are obtained by prior-by-prior Bayesian updating on all histories reachable by the trading strategy and pricing rule. Applying a similar argument as Proposition 2.3, we can show that $\delta_t = \delta_1$ for all $t > 1$ if. We show that trading strategy and pricing rule satisfy equilibrium conditions on all reachable histories given such a belief system.

First, the optimality of informed traders' strategies is straightforward given that $a_t = \underline{\pi}_t^{(1)}$ and $b_t = \bar{\pi}_t^{(0)}$. Second, given that informed traders' strategies, the market maker posts prices to meet zero profit conditions as follows:

$$\begin{aligned} a_t &= \max_{\pi_t \in \Pi_t} E^{\pi_t} [v|h_t, 0 < z_t \leq z^*], \\ &= \frac{\bar{\pi}_t(pq + \hat{\gamma}z^*)}{\bar{\pi}_t(pq + \hat{\gamma}z^*) + (1 - \bar{\pi}_t)[p(1 - q) + \hat{\gamma}z^*]}, \\ b_t &= \min_{\pi_t \in \Pi_t} E^{\pi_t} [v|h_t, -z^* \leq z_t < 0], \\ &= \frac{\underline{\pi}_t[p(1 - q) + \hat{\gamma}z^*]}{\bar{\pi}_t[p(1 - q) + \hat{\gamma}z^*] + (1 - \bar{\pi}_t)(pq + \hat{\gamma}z^*)}, \end{aligned}$$

where $\hat{\gamma} = (1 - \eta)/2\eta$. By equating $\underline{\pi}_t^{(1)}$ and $\bar{\pi}_t^{(0)}$ with $\max_{\pi_t \in \Pi_t} E^{\pi_t} [v|h_t, 0 < z_t \leq z^*]$ and $\min_{\pi_t \in \Pi_t} E^{\pi_t} [v|h_t, -z^* \leq z_t < 0]$, respectively, either of them yields that

$$\delta_t = \frac{q}{1 - q} \frac{p(1 - q) + \hat{\gamma}z^*}{pq + \hat{\gamma}z^*}. \quad (\text{A.3})$$

If

$$\frac{q}{1 - q} \frac{(1 - q) + \hat{\gamma}}{q + \hat{\gamma}} \leq \delta_t < \frac{q}{1 - q},$$

we can pick $z^* = 1$ and some $p \in (0, 1]$ to make equation (A.3) hold. This is because, given $z^* = 1$, its right-hand side equals $q/(1 - q)$ if $p = 0$, and it equals $\frac{q}{1 - q} \frac{(1 - q) + \hat{\gamma}}{q + \hat{\gamma}}$ if $p = 1$. If

$$1 < \delta_t < \frac{q}{1 - q} \frac{(1 - q) + \hat{\gamma}}{q + \hat{\gamma}},$$

we can pick some $z^* \in (0, 1)$ and $p = 1$ to make equation (A.3) hold. This is because, given $p = 1$, its right-hand side equals to $1 < \delta_t$ if $z^* = 0$ and equals $\frac{q}{1 - q} \frac{(1 - q) + \hat{\gamma}}{q + \hat{\gamma}}$ if $z^* = 1$.

Case (ii): Consider a belief system $\Pi_t = \Pi_1$ for all $t \geq 1$. It is consistent to the trading strategy and pricing rule. In each reachable history, the market maker learns nothing about the asset value since informed traders do not trade. Without informative signals, prior-by-prior Bayesian updating yields an exactly same set of posteriors as the set of priors. Then, $\delta_t = \delta_1 > q/(1 - q)$ implies that $\underline{\pi}_t^{(1)} > \bar{\pi}_t = a_t$ and $\bar{\pi}_t^{(0)} < \underline{\pi}_t = b_t$. It justifies the optimality of an informed trader's strategy. Thus, it simply set prices as follows to meet zero profit conditions,

$$\begin{aligned} a_t &= \max_{\pi_t \in \Pi_t} E^{\pi_t} [v|h_t, z_t \geq 0] = \bar{\pi}_t, \\ b_t &= \min_{\pi_t \in \Pi_t} E^{\pi_t} [v|h_t, z_t \leq 0] = \underline{\pi}_t, \end{aligned}$$

since observed order z_t is not informative. $\square$

Data availability

No data was used for the research described in the article.

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